Quillwave: Sharp Ripple Fields from 64 Sensors

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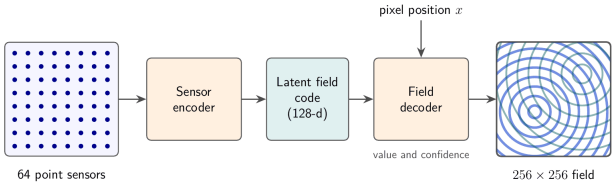


Figure 1: Quillwave pipeline.

Abstract

Quillwave reconstructs a full 256×256 ripple field from 64 point sensors. Ripple fields are procedurally generated sums of interfering circular waves: cheap to synthesise, but hard to interpolate, because the wave fronts between sensors are lost by smooth interpolation. Quillwave encodes the sensor readings into a 128-dimensional latent field code and decodes it with a coordinate-based field decoder. On RippleBench-Synthetic, a benchmark of 500 synthetic test fields generated for this work, Quillwave reaches 29.46 dB PSNR and 0.887 SSIM, 5.41 dB above radial-basis interpolation (the best baseline), which is a 71.2 % lower mean squared error. It is also 1.36× faster than radial-basis interpolation at inference. All results are on synthetic fields only.

1 Method

Let $s \in \mathbf{R}^{64}$ be the sensor readings at known positions p_i on a regular 8×8 grid. The sensor encoder E_ϕ maps the pairs (p_i, s_i) to a latent field code $z = E_\phi(s) \in \mathbf{R}^{128}$. The field decoder D_θ is coordinate-based: for any pixel position x it predicts a value and a confidence,

$$\big(\hat{f}(x),\, c(x)\big) = D_\theta(x, z). \tag{1}$$

Training minimises the mean squared error between $\hat{f}$ and the ground-truth field over all pixels, plus a term that asks $c(x)$ to rank pixels by their error. Quillwave-Lite removes the latent code and conditions the decoder on the raw readings directly. The full model has 1.2M parameters and is trained for 200 epochs.

2 Results

RippleBench-Synthetic fields are 256×256 sums of damped circular waves from up to four sources with random positions, wavelengths and phases. We use 2000 training fields and 500 test fields, and 64 sensors unless stated otherwise. We compare against linear interpolation and radial-basis interpolation from the same sensors, and against the Quillwave-Lite ablation. Learned models are trained 5 times with different seeds; we report the mean over the 5 training runs. Runtime is per field at inference.

Table 1: Reconstruction quality on RippleBench-Synthetic (64 sensors, 256×256 fields, mean over 5 training runs). Higher PSNR/SSIM is better; lower runtime is better.

Method	PSNR (dB) ↑	SSIM ↑	Runtime (ms) ↓
Linear interpolation	21.37	0.612	**0.8**
Radial-basis interpolation	24.05	0.701	6.4
Quillwave-Lite (ablation)	27.82	0.834	3.1
Quillwave (ours)	**29.46**	**0.887**	4.7

Table 1 gives the main result. Quillwave reaches 29.46 dB PSNR and 0.887 SSIM, 5.41 dB above radial-basis interpolation, the best baseline; in terms of mean squared error this is a 71.2 % reduction. At 4.7 ms per field it is 1.36× faster than radial-basis interpolation (6.4 ms), though slower than linear interpolation (0.8 ms). The ablation Quillwave-Lite reaches 27.82 dB, so the latent field code accounts for part of the gain.

3 Limitations

- **Baselines are not compute-matched.** Quillwave is trained for 3.5 GPU-hours, while the interpolation baselines need no training at all.
- **Synthetic fields only.** Every result is on RippleBench-Synthetic; no real sensor data was used, and behaviour on real measurements is unknown.
- **Confidence is not calibration.** The per-pixel confidence map that Quillwave outputs has not been checked against observed error rates, so it should not be read as a calibrated probability.

References

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